

TRACE AND TRANSFORM

Arduino Uno Based Embedded System Trackathon

Organized by: EES in collaboration with Arduino
Problem Statement v1.1

Prizes & Incentives

- **Training Opportunity** at Qualcomm
- **Cash Prizes**
- **ECA Credits**
- **Certifications and LORs**

1 PROBLEM OVERVIEW

Design and build an autonomous line-following robot using Arduino Uno that can navigate a complex track with multiple challenges including line gaps, color changes, elevation changes, obstacles, variable speed zones, and loose terrain. The robot must be intelligent enough to handle real-world scenarios beyond basic line following.

2 OBJECTIVES

- Design and fabricate a custom chassis without using pre-made kits
- Implement robust line detection and following algorithms
- Handle dynamic track conditions (gaps, color changes, obstacles, sand)
- Optimize speed based on track geometry
- Navigate elevation changes and inclines
- Demonstrate creativity in electronic and mechanical design

3 HARDWARE SPECIFICATIONS

3.1 Mandatory Components

Primary Controller: Arduino Uno (provided)

Power Supply: Any battery configuration (voltage regulation required)

3.2 Organizer-Provided Kit

The following components will be provided in the starter kit:

- IR sensors
- 4 DC motors and motor drivers (L298N)
- 4 wheels (you're not required to use them all)
- Jumper wires, connectors, and basic electronic components

3.3 Student-Provided Components

- **Chassis:** Must be custom-built by students using materials of their choice (wood, acrylic, 3D printed parts, metal, cardboard, etc.)
- **Creative Materials Encouraged:** Innovative use of everyday materials is welcomed and will be rewarded in evaluation

Note: Component requests must be justified based on the robot's design approach. Availability is subject to organizer discretion and stock levels.

3.4 Robot Specifications

Parameter	Specification
Maximum Width	20 cm (must fit within track boundaries)
Maximum Length	30 cm
Maximum Height	15 cm (excluding antenna/flags)
Number of Wheels	Student's choice (2, 3, 4, or more)
Number of Motors	Student's choice
Weight	No restriction (consider stability)

4 TRACK SPECIFICATIONS

4.1 Basic Track Properties

Parameter	Specification
Track Width (Arena)	30 cm (robot must stay within boundaries)
Line Width	2-5 cm (may vary at different sections)
Line Color(s)	Black, Red, Blue (on white background)
Surface Material	Matte white paper/cardboard/vinyl/gravel

4.2 Track Features

- **Straight sections:** For speed optimization
- **Gentle curves:** 90° and 45° turns
- **Sharp curves:** S-curves and hairpin bends
- **Variable line widths:** 2cm to 5cm transitions

5 CHALLENGE SPECIFICATIONS

5.1 Challenge 1: Line Gaps

Description: The line will have intentional gaps where the line discontinues and resumes after a distance.

Specifications:

- Gap length: 5-15 cm
- Number of gaps: 2-4 throughout the track
- Robot must predict line continuation and move forward
- Robot must resume line following after crossing the gap

Possible Edge Cases:

- Gap at a curve: Robot must maintain direction during gap
- Variable gap lengths in same track
- False gap detection (shadows, reflections must not be mistaken as gaps)

5.2 Challenge 2: Color Change

Description: The track line will change colors at certain sections (black → red → blue → black).

Specifications:

- Colors used: Black, Red, Blue
- Transitions can be abrupt or gradual (blended zone of 2-5 cm)

- Robot must adapt to color changes in real-time

Edge Cases:

- Color transition during a turn
- Lighting variations affecting color detection
- Similar shades that may confuse sensors

5.3 Challenge 3: Line Split with Color-Based Decision

Description: A single line will split into two lines of different colors. The robot must choose a path based on a predefined condition.

Specifications:

- Split configuration: One incoming line → Two outgoing lines (different colors)
- Example: Black line splits into Red (left) and Blue (right)
- Decision condition will be provided soon
- Split angle: 30° to 60° divergence

Possible Decision Conditions:

- "Always follow red line"
- "Follow blue line on first split, red on second"
- "Alternate colors at each split"

Edge Cases:

- Unequal divergence angles
- One path leading to immediate turn
- Multiple splits on the same track

5.4 Challenge 4: Elevation Changes and Inclines

Description: The track will include sections with elevation changes requiring the robot to navigate slopes and slanted surfaces.

Specifications:

- Uphill incline: 10° to 20° angle
- Downhill decline: 10° to 20° angle
- Minimum ramp length: 40 cm
- Ramp surface: Same material as track with anti-slip texture

Edge Cases:

- Maintaining line detection on slopes
- Sufficient torque for uphill movement
- Controlled descent on downhill sections
- Curve immediately after incline/decline

5.5 Challenge 5: Obstacles and Retracing

Description: Physical obstacles may block the path, requiring detection and backtracking capabilities.

Specifications:

- **Obstacle type:** Solid wall/barrier (cardboard/wood)
- **Obstacle height:** 15-30 cm
- **Obstacle width:** Completely blocks path (track width)
- **Robot must detect obstacle before collision**
- **Robot must reverse/retrace to previous junction or decision point**
- **Robot must choose alternate path if available**

Required Behaviors:

1. **Detection:** Identify obstacle from safe distance
2. **Stop:** Halt before collision
3. **Decision:** Determine if alternate path exists
4. **Retrace:** Move backward while maintaining line contact
5. **Recovery:** Navigate to alternate route or wait for manual intervention

Edge Cases:

- **Obstacle after a gap or curve**
- **False positive detection** (normal track features)
- **Maintaining line contact during reverse motion**

5.6 Challenge 6: The Sand Trap (Loose Terrain)

Description: The robot must navigate through a section of the track covered in loose material (sand or fine gravel). The line will remain visible through the material.

Specifications:

- **Terrain Material:** Fine sand or small gravel (particle size 2mm - 5mm)
- **Layer Depth:** Sparse coverage; the track surface is uneven but the line is not fully buried.
- **Section Length:** 30 cm to 60 cm.
- **Visibility:** The contrast between the line and the background will be preserved, but sensors may need to account for noise/dust.

Key Difficulties:

- **Traction Loss:** Wheels may slip, causing the robot to drift off the line.
- **Sensor Interference:** Dust accumulation on sensors or uneven scattering of IR light due to gravel.

- **Torque Management:** Motors must provide consistent power to push through resistance without stalling.

Edge Cases:

- Gravel getting stuck in wheel mechanisms.
- Sudden transition from smooth paper to rough sand.
- Robot "digging itself in" by spinning wheels too fast in one spot.

6 RULES AND CONSTRAINTS

6.1 General Rules

1. Robot must be **fully autonomous** after starting. No remote control or manual intervention allowed during run.
2. Arduino Uno must be the **primary controller**. Additional microcontrollers are not permitted.
3. Robot must **start from a designated start point** and follow the line to the finish.
4. Teams will have **multiple attempts** (typically 2 trials) with the best run counting toward final score.
5. Robots that **lose the line** for more than 5 seconds will be stopped and penalized.
6. Physical dimensions must be verified before each run.
7. Safety is paramount - robots with exposed high-voltage components or dangerous mechanisms will be disqualified.

6.2 Chassis Constraints

Mandatory: Chassis must be student-built and cannot be:

- Pre-made robot chassis kits
- Purchased robot frames
- Borrowed complete chassis from previous projects

- **Allowed:** Raw materials (wood, acrylic, metal sheets, 3D printed parts, cardboard, PVC pipes, bottles etc.)
- We can provide chassis to students but it will carry a significant penalty.
- **Encouraged:** Creative and unconventional materials

6.3 Sensor and Electronics Constraints

- Sensors must be mounted within the robot's physical dimensions
- No external lighting sources to aid line detection
- All electronics must be powered by onboard batteries
- Wireless communication for control is prohibited

6.4 Safety Requirements

- All exposed electrical connections must be properly insulated
- Sharp edges must be covered or filed smooth
- Batteries must be secured to prevent movement during operation
- Emergency stop mechanism (physical button or tilt switch) is recommended

7 SUBMISSION REQUIREMENTS

7.1 Phase 1: Design Submission

Mandatory Submission Deadline: January 31, 2026

Teams must submit the following documents:

1. Chassis Design Documentation:

- CAD files (SolidWorks, Fusion 360, AutoCAD, or similar) OR
- Detailed hand-drawn technical drawings on paper with dimensions
- All dimensions clearly labeled
- Material specifications

2. Circuit Diagram:

- Complete wiring schematic
- Pin connections to Arduino Uno
- Power distribution circuit
- All component labels and values

3. Component Request Form:

- List of sensors/components needed from organizers
- Quantities required for each component
- Specifications for each component (model numbers, ratings)
- Justification for each requested component
- Alternative options if primary choice unavailable

Submission Format:

- Single PDF document
- File naming: TeamName_Design_24Jan2026.pdf
- Submit via [platform/email to be announced]
- Include team member names and contact information on first page

7.2 Phase 2: Procurement Update (First Week of February, 2026)

Organizers will review all component requests and provide:

- Approved component list for each team
- Component availability status
- Alternative suggestions for unavailable items
- Component distribution schedule
- Any modifications to BOM if needed

7.3 Phase 3: Primary Demo Run (February 11, 2026)

First Evaluation: February 11, 2026

Teams must present a working prototype:

- Robot must be substantially complete
- Basic line-following capability must be demonstrated
- At least 3 out of 6 challenges should be attempted
- Source code (work-in-progress version acceptable)

Primary Demo Run Details:

- **Purpose:** Establish baseline performance and create primary leaderboard
- **Attempts:** Each team gets 2 official runs
- **Scoring:** Partial scoring based on challenges completed
- **Feedback:** Organizers and mentors will provide improvement suggestions
- **Leaderboard:** Rankings will be displayed to motivate teams
- **Note:** This is NOT the final evaluation - teams have one week to improve

7.4 Phase 4: Final Judging (February 13, 2026)

Final Competition: February 13, 2026

Judged by Arduino Professionals and Professors

Teams must present their final, polished robot:

1. Final Robot Inspection:

- Dimension verification (25cm × 30cm × 25cm max)
- Safety inspection
- Weight check and stability assessment
- Aesthetic and build quality evaluation

2. Competition Runs:

- 20-minute calibration time before runs
- Best run counts toward final score
- Timed runs with live scoring

3. Final Code Submission:

- Well-documented, commented source code
- README with setup and calibration instructions
- Libraries and dependencies list

8 RESOURCES & SUPPORT ---

8.1 Event Support

- Arduino workshops (scheduled separately)
- Mentors from the Electrical hobby group will be assigned to each team.

8.2 Workshop Space

- Access to maker space, embedded lab etc. for construction